

Topology as a Computational Basis for ICL in First-Order CRNs

A concise account of the theory, experiments, audit, and current evidence

Topology worktree: [topology](#)
Evidence source commit: [ba0e1a69](#)
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Summary. For first-order CRNs with exponential input-dependent rates, topology is not just a container for trainable parameters. It determines the rooted-spanning-tree sums through which edge projections enter the steady state. The computational basis is therefore not the set of edge vectors $\{K_e\}$, but the set of tree-sum vectors

$$\Theta_T = \sum_{e \in T} K_e, \quad T \in \mathcal{T}_r(G), \quad r \in V.$$

The experiments support the scoped statement that, in the tested fixed-count regimes, topology-associated structural and functional variables explain residual novel-class ICL variation that raw trainable count does not explain. They do not yet give a universal scalar topology law.

Scope. Everything below is about first-order CRNs with exponential rate encoding. The matrix-tree theorem makes this case exact. Autocatalytic and WTA CRNs need separate topology theories.

[1] The starting question

The original CRN-ICL paper showed that a chemical network can do in-context learning without transformer attention. The input is a context/query vector

$$z = (z_1, \dots, z_{N_c}, z_q), \quad z_i \in \mathbb{R}^D,$$

and the target is the context position whose item matches the query. The true metric is novel-class ICL accuracy, not training accuracy or ordinary validation accuracy.

The paper’s mechanism was subspace projection. For each context position i , define a comparison region

$$\mathcal{M}_i = \{z : z_i \sim z_q\}.$$

ICL succeeds when inputs from different comparison branches produce steady states that the decoder separates.

The original degree-count story was broadly right:

$$n_{\text{req}} = 2N_c(N_c + 1)D, \quad n_g = N_n(N_n - 1)(N_c + 1)D$$

for fully connected first-order CRNs. But this count does not say where the degrees of freedom sit. The topology question was therefore

At fixed count, does the reaction/input topology change ICL?

[2] The exact first-order object

Let $G = (V, E)$ be a strongly connected directed first-order reaction graph. For edge $e : u \rightarrow v$, use exponential rate encoding

$$k_e(z) = \exp(b_e + K_e^\top z).$$

The first-order steady state is controlled exactly by the matrix-tree theorem:

$$\bar{C}_r(z) = \frac{\tau_r(z)}{\sum_s \tau_s(z)}, \quad \tau_r(z) = \sum_{T \in \mathcal{T}_r(G)} \prod_{e \in T} k_e(z).$$

Substitute the exponential form:

$$\prod_{e \in T} k_e(z) = \exp \left(\sum_{e \in T} b_e + \left(\sum_{e \in T} K_e \right)^\top z \right).$$

Set

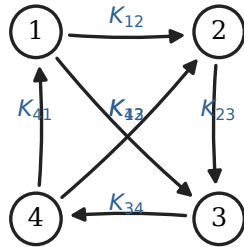
$$\beta_T = \sum_{e \in T} b_e, \quad \Theta_T = \sum_{e \in T} K_e.$$

Then

$$\tau_r(z) = \sum_{T \in \mathcal{T}_r(G)} \exp(\beta_T + \Theta_T^\top z).$$

This is the key point. A first-order CRN does not read out isolated edge projections $K_e^\top z$. It reads out log-sum-exp combinations of rooted-tree projections $\Theta_T^\top z$.

physical graph $G = (V, E)$



matrix-tree

computational basis

$$au_r(z) = \sum_{T \in \text{Trees}_r(G)} e^{\beta_T + \Theta_T^\top z}$$

$$\Theta_T = \sum_{e \in T} K_e$$

basis is $\{\Theta_T : T \in \text{Trees}_r(G), r \in V\}$
 not the isolated edge set $\{K_e : e \in E\}$

edges carry learned vectors K_e

Figure 1: Topology changes the projection basis. Edge vectors K_e are learned locally, but the steady state uses rooted tree-sum projections Θ_T .

[3] What topology changes

Stack tree incidences as row vectors $s_T \in \{0, 1\}^m$. Then

$$\Theta_T = K^\top s_T.$$

Since concentrations are normalized, common shifts of all tree scores cancel. The more relevant structural object is the tree-difference matrix

$$D_G = \begin{bmatrix} s_{T_1} - s_{T_0} \\ s_{T_2} - s_{T_0} \\ \vdots \end{bmatrix}.$$

A first count of relative projection capacity is

$$d_{\text{rel}}(G) = p \text{rank}(D_G), \quad p = (N_c + 1)D.$$

With an input mask $\Omega_{e\alpha}$, the capacity must be input-aware:

$$d_{\text{rel}}(G, \Omega) = \sum_{\alpha=1}^p \text{rank}(D_G \text{diag}(\Omega_{\cdot\alpha})).$$

But this is still only a rank proxy. ICL asks a sharper question:

Can the allowed tree-sum directions separate the comparison branches?

The capacity object should be closer to

$$\gamma^*(G, \Omega) = \max_{K, B} \min_{z \in \mathcal{B}} \left[q_{\ell^*(z)}(z) - \max_{\ell \neq \ell^*(z)} q_{\ell}(z) \right]$$

under norm constraints. In the tropical limit,

$$\log \tau_r(z) \approx \max_{T \in \mathcal{T}_r(G)} (\beta_T + \Theta_T^\top z),$$

so the theory becomes a rooted-tree-polytope normal-fan coverage problem.

The hierarchy that emerged is therefore

raw count < relative tree rank < masked tree geometry
 < branch-margin/tree-polytope geometry < trained organization < causal dependence.

[4] What was built and audited

The worktree now contains code and reports for arbitrary first-order directed topologies, input masks, tree metrics, branch-capacity probes, post-training tree diagnostics, ablations, motif extraction, and statistic-preserving causal scrambles.

The audit report `ICL/results/next_phase_stats/topology_goal_completion_audit.md` maps the goal to concrete artifacts. It records that the verifier passed for committed artifacts, that hard regimes have full result/model/mechanism/causal follow-ups, and that all headline claims are scoped to `test_novel_classes`. The audit also marks the branch-margin theory as partial by design: proxies exist, but there is not yet a final theorem.

The main source reports used here are:

topology_icl_complete_explanation.md
 next_phase_evidence_report.md/json
 mechanism_isolation_evidence.csv
 stat_preserving_causal_stratified
 degree_rewire_normal_fan_...

full narrative explanation
 fixed-count, motif, causal summaries
 controlled mechanism contrasts
 stratified causal scrambles
 exact-degree normal-fan pilot

[5] Fixed-count structural evidence

The original controlled regime fixed

$$N_n = 6, \quad m = 20, \quad \text{input-coupled count} = 200.$$

It used three physical backbones, 16 selected masks/topologies per backbone, five training seeds each, 240 total runs, and 48 topology/mask groups.

At group level, with target equal to mean novel-class ICL accuracy, the leave-one-out values were

model	group LOO R^2	interpretation
raw count	-0.043	no explanatory power at fixed count
raw count + d_{rel}	0.145	relative rank helps
masked tree geometry	0.189	input-aware geometry helps
tree geometry	0.409	strongest structural summary here

The corresponding hard $N_n = 5, m = 12, N_c = 3, D = 2$ regime showed a different but still positive pattern: raw count and d_{rel} alone were both -0.190, tree geometry was 0.324, and masked tree geometry was 0.447.

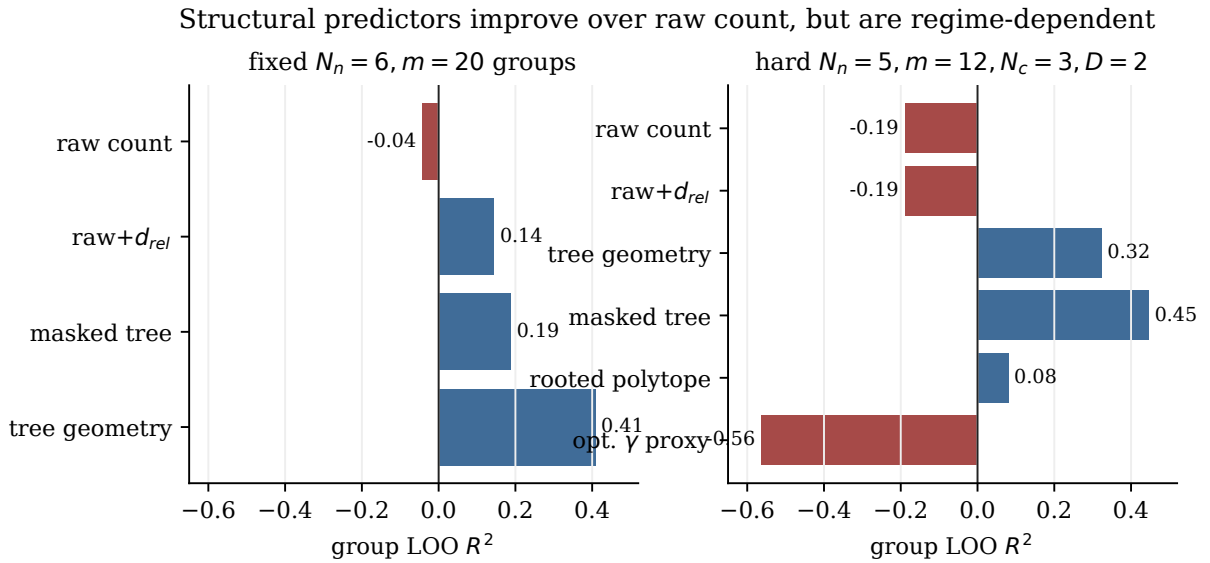


Figure 2: Structural predictors in two fixed-count regimes. Raw count is uninformative once fixed. Tree geometry and masked tree geometry explain residual group-level variation, but the best predictor depends on regime.

The safe conclusion is not that d_{rel} is the law. It is that topology-associated variables explain residual ICL variation that raw count cannot explain in the tested regimes.

[6] Why input masks matter

Physical topology and input-encoding topology are distinct. The physical graph fixes the rooted trees:

$$G_{\text{phys}} \mapsto \mathcal{T}_r(G).$$

The input mask fixes which coordinates can enter which edge vectors:

$$\Omega_{e\alpha} = 0 \quad \Rightarrow \quad K_{e\alpha} = 0.$$

The real capacity object is therefore $\gamma^*(G, \Omega)$, not only $\gamma^*(G)$.

The mechanism-isolation file shows fixed-graph, fixed-count, fixed- d_{rel} cases where masks still change ICL. On the fixed random physical graph, changing edge-load structure gave +6.96 mean ICL points, while changing coordinate-load heterogeneity gave −16.48 mean ICL points. On the fixed cycle graph, the corresponding deltas were −6.32 and +2.12 points; on the fixed hub graph they were +3.44 and −1.52 points. The sign is not universal, but the effect exists.

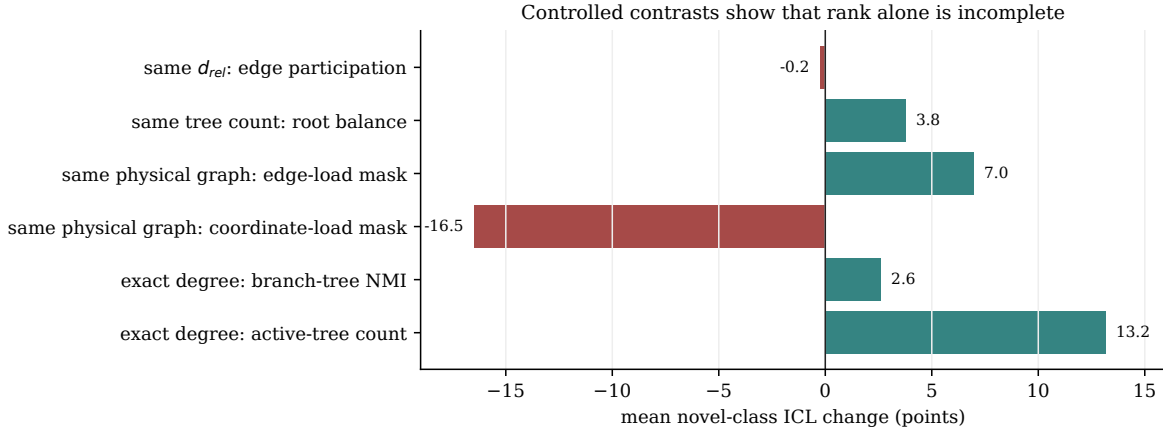


Figure 3: Controlled contrasts. Some coarse structural metrics are not monotone drivers, and fixed-graph input masks can still change ICL. This is why d_{rel} alone is incomplete.

[7] Post-training mechanism evidence

Pre-training topology asks what the graph could use. Post-training diagnostics ask what the trained model did use. The strongest diagnostic class was projection/root/tree organization around com-

parison branches. Projection-alignment summaries gave

$$\begin{aligned} R_{\text{LOO}}^2 &= 0.740 && \text{at run level,} \\ R_{\text{LOO}}^2 &= 0.809 && \text{at topology-mean level,} \\ R_{\text{LOO}}^2 &= 0.599 && \text{for topology-best seed.} \end{aligned}$$

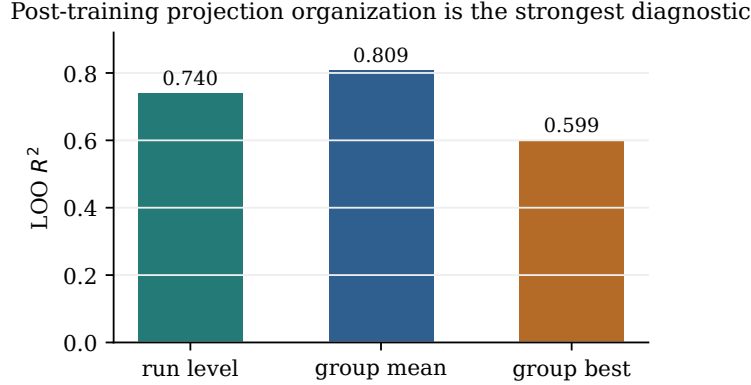


Figure 4: Post-training projection alignment is much more explanatory than coarse pre-training count. This is mechanism evidence, not an independent capacity theorem.

This means high-ICL models are not merely high-count systems. They organize the available tree-sum geometry so that comparison branches produce decodable steady states. The active-tree and margin diagnostics are close to the learned mechanism, so they should be read as mechanistic explanations rather than clean pre-training predictors.

[8] Causal scrambles

The causal interventions are the strongest evidence that the learned alignment matters. Two statistic-preserving scrambles were central:

branch-alignment scramble	preserves graph, root tree counts, d_{rel} , and input density
projection scramble	also preserves input support and per-edge projection row norms

They break branch/projection/tree organization while preserving many coarse statistics.

In the stratified hard-regime pilot, high-ICL runs dropped by about -96 points under branch-alignment scrambling and about -64 to -68 points under projection scrambling. Low and mid runs also collapsed substantially.

This pushes the evidence beyond correlation. It says the trained CRNs rely on the organized relation between input branches, learned edge projections, and root/tree structure.

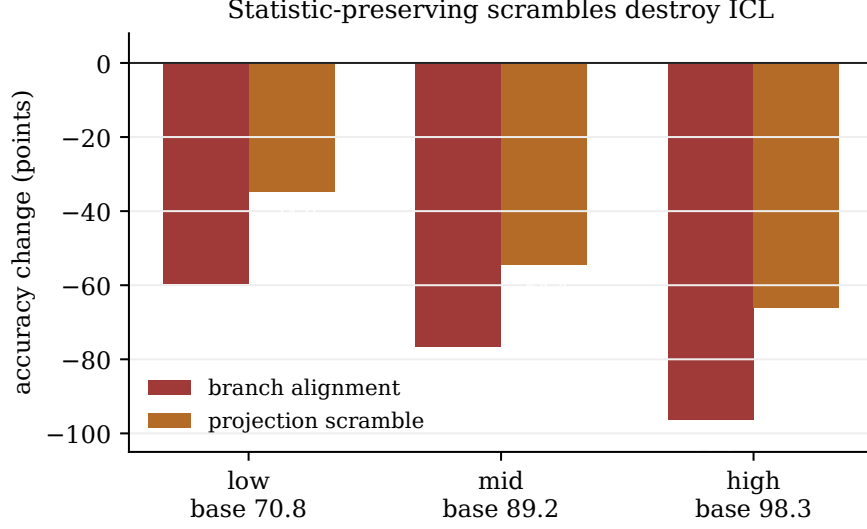


Figure 5: Statistic-preserving scrambles in low, mid, and high baseline buckets. Destroying branch/projection/tree alignment sharply reduces novel-class ICL.

[9] Hard regimes and the normal-fan pilot

Three hard regimes were added: $n4_m6_N3_D2$, $n5_m8_N3_D2$, and $n5_m12_N3_D2$, each with 60 trained runs plus mechanism and causal follow-ups. The results are mixed in the useful sense. Topology signal depends on regime.

In $n5_m12_N3_D2$, masked tree geometry had $LOO R^2 = 0.447$ and positive held-out-family performance. In the sparser $n5_m8_N3_D2$ regime, within-sample bootstrap improvements were more positive than held-out-family prediction. This suggests that topology may have sparse, intermediate, and redundant regimes rather than one monotone scalar rule.

The exact-degree normal-fan pilot fixed

$$N_n = 5, \quad m = 12, \quad N_c = 3, \quad D = 2, \quad \text{in/out degree sequence} = [3, 2, 2, 3, 2], \quad d_{\text{rel}} = 88.$$

It varied tree-polytope/normal-fan diagnostics through directed degree-preserving rewires. Only four topology groups were trained, so this is not yet a statistical result. It is a constructive design showing that topology can vary beyond degree sequence and d_{rel} .

The descriptive $n = 4$ correlations were $r = 0.869$ for active-tree count and $r = 0.691$ for branch-tree NMI with mean ICL. These numbers are promising design feedback, not proof.

[10] Motifs and matched controls

Essential physical subgraphs retrained better than sparse input masks under the extraction protocol. But matched controls softened the stronger interpretation. Degree-rewired and random strongly connected controls often matched or exceeded the extracted motifs.

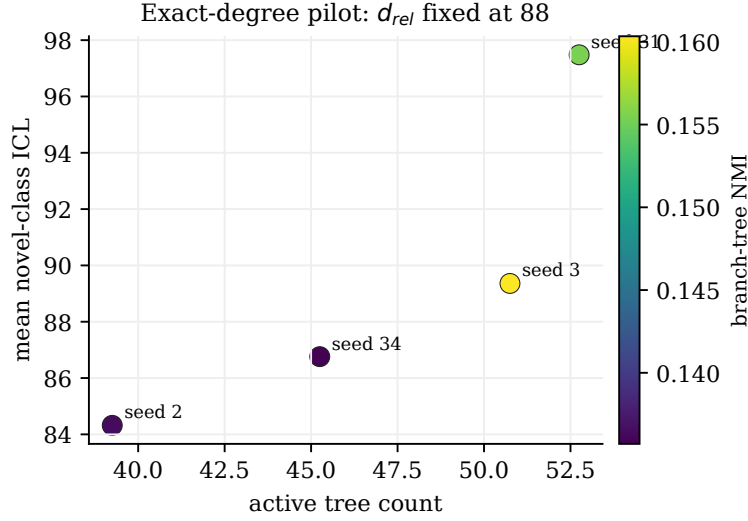


Figure 6: Exact-degree pilot. All points have the same degree sequence and $d_{\text{rel}} = 88$. Active tree count and branch-tree NMI vary across rewires, and mean ICL varies with them in this small $n = 4$ pilot.

The corrected interpretation is:

small physical subgraphs can support ICL, but the extracted motifs are not uniquely optimal.

This is compatible with the tree-sum view: many different graphs can provide usable branch-covering tree geometry.

[11] The gamma-star attempt

The project implemented γ^* -style capacity probes because d_{rel} is too coarse. This was the right theoretical direction, but the current optimized proxy did not outperform tree geometry. In the hard $n5_m12_N3_D2$ regression, optimized γ capacity had LOO $R^2 = -0.564$, while tree geometry had 0.324 and masked tree geometry had 0.447.

This is an informative failure. The surrogate likely optimizes smooth average behavior more than the true worst-branch margin. ICL is closer to a worst-branch problem:

$$\max_{K, B} \min_{b \in B} \text{margin}_b(K, B; G, \Omega).$$

A good capacity theory should target low-percentile or worst-branch separation under norm constraints.

[12] What has been achieved

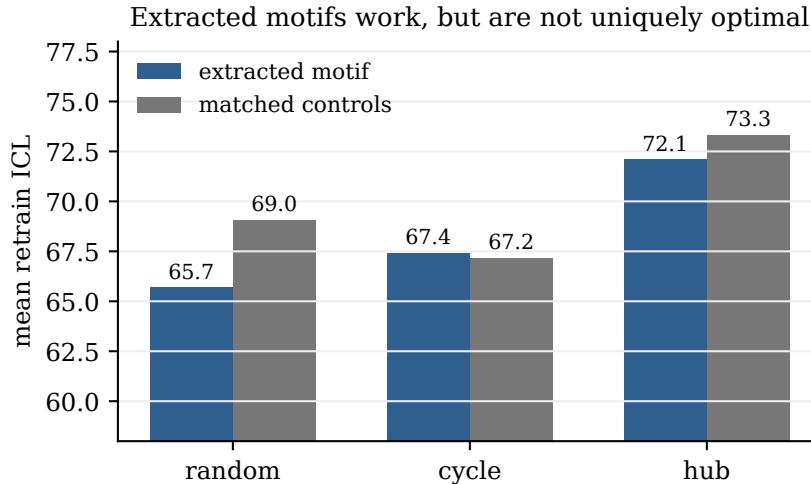


Figure 7: Extracted physical motifs support ICL, but matched controls are competitive. The evidence supports a family of feasible physical substrates, not one unique motif.

Before this direction, the paper had:

CRNs can do ICL by subspace projection, and raw learned degrees of freedom are important.

After this direction, the first-order story is sharper:

raw degrees are useful only insofar as topology and masks turn them into relative tree-sum projections that cover comparison branches.

The concrete accomplishments are:

Layer	What was established
Theory	The exact first-order computational basis is the rooted tree-sum basis $\{\Theta_T\}$, not the isolated edge basis $\{K_e\}$.
Structural evidence	Fixed-count regimes show residual ICL variation explained by tree and masked-tree geometry beyond raw count.
Input topology	Fixed physical graph/count/ d_{rel} masks can still differ sharply, so the object is $\gamma^*(G, \Omega)$.
Mechanism	Successful trained models organize projection/root/tree features around comparison branches.
Causality	Statistic-preserving scrambles that break alignment sharply reduce ICL.
Limits	d_{rel} is not a capacity law; current γ probes are not final; motifs are not unique; nonlinear CRNs are out of scope.

The most defensible statement is:

In the tested first-order fixed-count regimes, topology-associated structural and functional variables explain residual novel-class ICL variation that raw trainable count does not explain. The strongest mechanism evidence is that trained models depend on branch/projection/tree alignment.

[13] What should come next

The next phase should not be a loose sweep. It should scale the exact-degree/normal-fan design:

fix N_n, m, N_c, D , input count, degree sequence, and d_{rel} ,

vary tree-polytope and normal-fan diagnostics, then train enough groups for grouped inference.

Then compare raw count, d_{rel} , masked tree geometry, normal-fan diagnostics, and an improved worst-branch $\gamma^*(G, \Omega)$ probe.

The theoretical target is:

derive a norm-constrained tree-polytope branch-margin capacity that predicts whether (G, Ω) can cover ICL branches.

That would turn the current empirical and mechanistic evidence into a true first-order topology theory.